1) a)

$y = 3x + 5$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + (3)^2 = 10$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{10}$

$\Rightarrow y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = (3x + 5)\sqrt{10}$

$S = 2\pi \int_0^2 y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\sqrt{10}\pi \int_0^2 (3x + 5)\,\mathrm{d}x$

$= 2\sqrt{10}\pi \left[\frac{3}{2}x^2 + 5x\right]_0^2$

$= 2\sqrt{10}\pi[6 + 10 - 0] = 32\sqrt{10}\pi$ units^2

1) b)

$y = 2\sqrt{x} = 2x^{\frac{1}{2}}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = x^{-\frac{1}{2}}$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + \left(x^{-\frac{1}{2}}\right)^2 = 1 + \frac{1}{x}$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1 + \frac{1}{x}}$

$\Rightarrow y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = 2\sqrt{x}\sqrt{1 + \frac{1}{x}} = 2\sqrt{x}\sqrt{\frac{x+1}{x}} = 2\sqrt{x + 1}$

$S = 2\pi \int_0^3 y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi \int_0^3 2(x + 1)^{\frac{1}{2}}\,\mathrm{d}x$

$= 2\pi \left[\frac{4}{3}(x + 1)^{\frac{3}{2}}\right]_0^3$

$= 2\pi \left[\frac{32}{3} - \frac{4}{3}\right] = \frac{56\pi}{3}$ units^2

1) c)

$y = \cosh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh x$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + \sinh^2 x = \cosh^2 x$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{\cosh^2 x} = \cosh x$

$\Rightarrow y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \cosh x \cosh x = \cosh^2 x = \frac{1}{2}+\frac{1}{2}\cosh 2x$

$S = 2\pi\int_0^1 y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi\int_0^1\left(\frac{1}{2}+\frac{1}{2}\cosh 2x\right)\mathrm{d}x = \pi\int_0^1(1+\cosh 2x)\,\mathrm{d}x$

$= \pi\left[x+\frac{1}{2}\sinh 2x\right]_0^1$

$= \pi\left(1+\frac{1}{2}\sinh 2\right) = \frac{\pi}{2}(2+\sinh 2) = \frac{\pi}{2}\left(2+\frac{1}{2}\mathrm{e}^2-\frac{1}{2}\mathrm{e}^{-2}\right) = \frac{\pi}{4}(4+\mathrm{e}^2-\mathrm{e}^{-2})$ units^2

1) d)

$y = \frac{1}{2\sqrt{3}}x^{\frac{1}{2}}(4-x) = \frac{2}{\sqrt{3}}x^{\frac{1}{2}} - \frac{1}{2\sqrt{3}}x^{\frac{3}{2}}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{3}}x^{-\frac{1}{2}} - \frac{3}{4\sqrt{3}}x^{\frac{1}{2}}$

$\therefore 1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1+\left(\frac{1}{\sqrt{3x}}-\frac{3\sqrt{x}}{4\sqrt{3}}\right)^2 = 1+\frac{1}{3x}-\frac{1}{2}+\frac{3x}{16} = \frac{1}{2}+\frac{1}{3x}+\frac{3x}{16} = \frac{9x^2+24x+16}{48x} = \frac{(3x+4)^2}{48x}$

$\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{\frac{(3x+4)^2}{48x}} = \frac{3x+4}{4\sqrt{3}\sqrt{x}}$

$\Rightarrow y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \left(\frac{2\sqrt{x}}{\sqrt{3}}-\frac{x\sqrt{x}}{2\sqrt{3}}\right)\frac{3x+4}{4\sqrt{3}\sqrt{x}} = \frac{2\sqrt{x}}{\sqrt{3}}\left(\frac{3x+4}{4\sqrt{3}\sqrt{x}}\right)-\frac{x\sqrt{x}}{2\sqrt{3}}\left(\frac{3x+4}{4\sqrt{3}\sqrt{x}}\right) = \frac{3x+4}{6}-\frac{3x^2+4x}{24}$

$= \frac{1}{24}(-3x^2+8x+16)$

$S = 2\pi\int_1^4 y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = \frac{2\pi}{24}\int_1^4(-3x^2+8x+16)\,\mathrm{d}x = \frac{\pi}{12}[-x^3+4x^2+16x]_1^4$

$= \frac{\pi}{12}[(-64+64+64)-(-1+4+16)] = \frac{45\pi}{12} = \frac{15\pi}{4}$ units^2

2)

$y = ax$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\therefore 1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1+a^2$

$\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1+a^2}$

$\Rightarrow y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = ax\sqrt{1+a^2}$

$S = 2\pi\int_0^k y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi\int_0^k ax\sqrt{1+a^2}\,\mathrm{d}x$

$= 2\pi a\sqrt{1+a^2}\left[\frac{1}{2}x^2\right]_0^k$

$= \pi a k^2\sqrt{1+a^2}$ units^2

3) a)

$y = (x+16)^{\frac{1}{2}}$

$\frac{dy}{dx} = \frac{1}{2}(x+16)^{-\frac{1}{2}}$

$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(\frac{1}{2}(x+16)^{-\frac{1}{2}}\right)^2 = 1 + \frac{1}{4(x+16)} = \frac{4(x+16)+1}{4(x+16)} = \frac{4x+65}{4(x+16)}$

$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{4x+65}{2(x+16)}} = \frac{\sqrt{4x+65}}{2\sqrt{x+16}}$

$\Rightarrow y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{x+16}\,\frac{\sqrt{4x+65}}{2\sqrt{x+16}} = \frac{\sqrt{4x+65}}{2}$

$S = 2\pi \int_0^6 y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^6 \left(\frac{\sqrt{4x+65}}{2}\right)dx$

$= \pi \int_0^6 (4x+65)^{\frac{1}{2}}\,dx$

$= \pi \left[\frac{1}{6}(4x+65)^{\frac{3}{2}}\right]_0^6$

$= \pi \left[\frac{1}{6}(89)^{\frac{3}{2}} - \frac{1}{6}(65)^{\frac{3}{2}}\right] = 165.2$ cm^2 (4 SF)

3) b)

Cost: $20 \times 165.2 \times 0.03 =$£99 (to the nearest £1)

4)

$y = \sqrt{x+1} = (x+1)^{\frac{1}{2}}$

$\frac{dy}{dx} = \frac{1}{2}(x+1)^{-\frac{1}{2}} = \frac{1}{2\sqrt{x+1}}$

$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(\frac{1}{2\sqrt{x+1}}\right)^2 = 1 + \frac{1}{4(x+1)} = \frac{4(x+1)+1}{4(x+1)} = \frac{4x+5}{4(x+1)}$

$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{4x+5}{4(x+1)}}$

$\Rightarrow y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{x+1}\sqrt{\frac{4x+5}{4(x+1)}} = \frac{1}{2}\sqrt{4x+5}$

$S = 2\pi \int_0^k y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^k \frac{1}{2}\sqrt{4x+5}\,dx$

$= \pi \int_0^k (4x+5)^{\frac{1}{2}}\,dx$

$= \pi \left[\frac{1}{6}(4x+5)^{\frac{3}{2}}\right]_0^k$

$= \pi\left[\frac{1}{6}(4k+5)^{\frac{3}{2}} - \frac{5\sqrt{5}}{6}\right]$

But since $\frac{\pi}{6}\left(13\sqrt{13} - 5\sqrt{5}\right) = \pi\left[(4k+5)^{\frac{1}{2}} - \frac{5\sqrt{5}}{6}\right]$ then $\frac{1}{6}(4k+5)^{\frac{3}{2}} = \frac{13\sqrt{13}}{6}$

$(4k+5)^3 = 2197 \Rightarrow 4k+5 = 13$

$\therefore k = 2$

5)

When $y = 0$, so $0 = k - x^2 \Rightarrow x = \sqrt{k}$

$y = (k-x^2)^{\frac{1}{2}}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -x(k-x^2)^{-\frac{1}{2}} = -\frac{x}{\sqrt{k-x^2}}$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + \left(-\frac{x}{\sqrt{k-x^2}}\right)^2 = 1 + \frac{x^2}{k-x^2} = \frac{k-x^2+x^2}{k-x^2} = \frac{k}{k-x^2}$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{\frac{k}{k-x^2}}$

$\Rightarrow y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{k-x^2}\sqrt{\frac{k}{k-x^2}} = \sqrt{k-x^2}\frac{\sqrt{k}}{\sqrt{k-x^2}} = \sqrt{k}$

$S = 2\pi\int_0^{\sqrt{k}} y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi\int_0^{\sqrt{k}} \sqrt{k}\,\mathrm{d}x$

$= 2\pi\left[\sqrt{k}x\right]_0^{\sqrt{k}}$

$= 2\pi[k-0] = 2\pi k$

$\therefore 2 \times 2\pi k = 4\pi k$ units^2

6) a)

$\int \sqrt{x^2+1}\,\mathrm{d}x$

Let $x = \sinh t \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \cosh t \Rightarrow \mathrm{d}x = \cosh t\,\mathrm{d}t$

$\therefore \int \sqrt{x^2+1}\,\mathrm{d}x$

$= \int \sqrt{\sinh^2 t - 1} \times \cosh t\,\mathrm{d}t$

$= \int \sqrt{\cosh^2 t} \times \cosh t\,\mathrm{d}t$

$= \int \cosh t \times \cosh t\,\mathrm{d}t$

$= \int \cosh^2 t\,\mathrm{d}t$

$\cosh 2t \equiv 2\cosh^2 t - 1 \Rightarrow \cosh^2 t \equiv \frac{1}{2}(\cosh 2t + 1)$

$\therefore \int \cosh^2 t\,\mathrm{d}u = \int\left(\frac{1}{2}(\cosh 2t + 1)\right)\mathrm{d}t$

$= \frac{1}{2}\int(\cosh 2t + 1)\,\mathrm{d}t$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2t + t\right) + c$

$= \frac{1}{4}\sinh 2t + \frac{1}{2}t + c$

$\sinh 2t = 2\sinh t \cosh t = 2\sinh t \times \sqrt{\sinh^2 t + 1} = 2x\sqrt{x^2+1}$

$\therefore \int \sqrt{x^2+1}\,\mathrm{d}x = \frac{1}{4} \times 2x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c$

$= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln\left(x + \sqrt{x^2+1}\right) + c$

6) b)

$y = \mathrm{e}^{2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\mathrm{e}^{2x}$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + (2\mathrm{e}^{2x})^2 = 1 + 4\mathrm{e}^{4x}$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1 + 4\mathrm{e}^{4x}}$

$\Rightarrow y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \mathrm{e}^{2x}\sqrt{1 + 4\mathrm{e}^{4x}}$

$S = 2\pi \int_0^{\ln 2} y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi \int_0^{\ln 2} \mathrm{e}^{2x}\sqrt{1 + 4\mathrm{e}^{4x}}\,\mathrm{d}x$

Let $u = 2\mathrm{e}^{2x} \Rightarrow u^2 = 4\mathrm{e}^{4x}$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = 4\mathrm{e}^{2x}$

When $x = 0$, $u = 2$, when $x = \ln 2$, $u = 8$

$2\pi \int_0^{\ln 2} \mathrm{e}^{2x}\sqrt{1 + 4\mathrm{e}^{4x}}\,\mathrm{d}x = 2\pi \int_2^8 \frac{1}{4}\sqrt{1+u^2}\,\mathrm{d}u$

$= \frac{\pi}{2}\int_2^8 \sqrt{1+u^2}\,\mathrm{d}u$

$= \frac{\pi}{2}\left[\frac{1}{2}\operatorname{arsinh} u + \frac{1}{4}\sinh(2\operatorname{arsinh} u)\right]_2^8$

$= \frac{\pi}{2}\left[\left(\frac{1}{2}\ln\left(8 + \sqrt{65}\right) + 4\sqrt{65}\right) - \left(\frac{1}{2}\ln\left(2 + \sqrt{5}\right) + \sqrt{5}\right)\right] = 48.2 \text{ units}^2 \text{ (3 SF)}$

7) a)

$y = \frac{1}{3}\cos(3x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin(3x)$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + (-\sin(3x))^2 = 1 + \sin^2(3x)$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1 + \sin^2(3x)}$

$\Rightarrow y\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \frac{1}{3}\cos(3x)\sqrt{1 + \sin^2(3x)}$

$S = 2\pi \int_0^{\frac{\pi}{6}} y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi \int_0^{\frac{\pi}{6}} \frac{1}{3}\cos(3x)\sqrt{1+\sin^2(3x)}\,\mathrm{d}x$

$= \frac{2\pi}{3}\int_0^{\frac{\pi}{6}} \cos(3x)\sqrt{1+\sin^2(3x)}\,\mathrm{d}x$ as req.

7) b)

Let $u = \sin(3x)$ so $\frac{\mathrm{d}u}{\mathrm{d}x} = 3\cos(3x)$

When $x = 0$, $u = 0$, when $x = \frac{\pi}{6}$, $u = 1$

$\frac{2\pi}{3}\int_0^{\frac{\pi}{6}} \cos(3x)\sqrt{1+\sin^2(3x)}\,\mathrm{d}x = \frac{2\pi}{3}\int_0^1 \frac{1}{3}\sqrt{1+u^2}\,\mathrm{d}u$

Now consider:

$\int \sqrt{u^2+1}\,\mathrm{d}x$

Let $u = \sinh t \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}t} = \cosh t \Rightarrow \mathrm{d}u = \cosh t\,\mathrm{d}t$

$\therefore \int \sqrt{u^2+1}\,\mathrm{d}u$

$= \int \sqrt{\sinh^2 t - 1} \times \cosh t\,\mathrm{d}t$

$= \int \sqrt{\cosh^2 t} \times \cosh t\,\mathrm{d}t$

$= \int \cosh t \times \cosh t\,\mathrm{d}t$

$= \int \cosh^2 t\,\mathrm{d}t$

$\cosh 2t \equiv 2\cosh^2 t - 1 \Rightarrow \cosh^2 t \equiv \frac{1}{2}(\cosh 2t + 1)$

$\therefore \int \cosh^2 t\,\mathrm{d}u = \int \left(\frac{1}{2}(\cosh 2t + 1)\right)\mathrm{d}t$

$= \frac{1}{2}\int (\cosh 2t + 1)\,\mathrm{d}t$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2t + t\right) + c$

$= \frac{1}{4}\sinh 2t + \frac{1}{2}t + c$

$\sinh 2t = 2\sinh t \cosh t = 2\sinh t \times \sqrt{\sinh^2 t + 1} = 2x\sqrt{x^2+1}$

$\therefore \int \sqrt{u^2+1}\,\mathrm{d}u = \frac{1}{4} \times 2u\sqrt{u^2+1} + \frac{1}{2}\operatorname{arsinh} u + c$

$= \frac{1}{2}u\sqrt{u^2+1} + \frac{1}{2}\ln\left(u + \sqrt{u^2+1}\right) + c$

So that

$\frac{2\pi}{3}\int_0^{\frac{\pi}{6}} \cos(3x)\sqrt{1+\sin^2(3x)}\,\mathrm{d}x = \frac{2\pi}{9}\left[\frac{1}{2}u\sqrt{u^2+1} + \frac{1}{2}\ln\left(u+\sqrt{u^2+1}\right)\right]_0^1$

$= \frac{2\pi}{9}\left[\left(\frac{1}{2}\sqrt{2} + \frac{1}{2}\ln\left(1+\sqrt{2}\right)\right) - (0+0)\right]$

$= \left(\frac{\pi\sqrt{2}}{9} + \frac{\pi}{9}\ln\left(1+\sqrt{2}\right)\right)$ units^2

8) a)

$y = \frac{\sqrt{7}}{3}(x+1)$

$\frac{dy}{dx} = \frac{\sqrt{7}}{3}$

$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(\frac{\sqrt{7}}{3}\right)^2 = \frac{16}{9}$

$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{16}{9}} = \frac{4}{3}$

$\Rightarrow x\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{4}{3}x$

$S = 2\pi \int_0^5 x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = \frac{8\pi}{3}\int_2^3 x\,dx$

$= \frac{8\pi}{3}\left[\frac{1}{2}x^2\right]_2^3 = \frac{8\pi}{3}\left[\frac{9}{2} - 2\right] = \frac{20\pi}{3}$ units^2

8) b)

$y = x^2$

$\frac{dy}{dx} = 2x$

$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + (2x)^2 = 1 + 4x^2$

$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + 4x^2}$

$\Rightarrow x\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x\sqrt{1 + 4x^2}$

$S = 2\pi \int_1^2 x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_1^2 x\sqrt{1 + 4x^2}\,dx = 2\pi \int_1^2 x(1 + 4x^2)^{\frac{1}{2}}\,dx$

$= 2\pi\left[\frac{1}{12}(1 + 4x^2)^{\frac{3}{2}}\right]_1^2 = 2\pi\left[\frac{17\sqrt{17}}{12} - \frac{5\sqrt{5}}{12}\right] = \frac{\pi}{6}\left(17\sqrt{17} - 5\sqrt{5}\right)$ units^2

8) c)

$y = \frac{2}{3}(x+3)^{\frac{3}{2}}$

$\frac{dy}{dx} = (x+3)^{\frac{1}{2}}$

$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left((x+3)^{\frac{1}{2}}\right)^2 = 1 + x + 3 = x + 4$

$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{x+4}$

$\Rightarrow x\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x\sqrt{x+4}$

$S = 2\pi \int_0^5 x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^5 x\sqrt{x+4}\,dx$

Let $u = x + 4$ so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$

When $x = 0$, $u = 4$, when $x = 5$, $u = 9$

$2\pi \int_0^5 x\sqrt{x+4}\,\mathrm{d}x = 2\pi \int_4^9 (u-4)u^{\frac{1}{2}}\,\mathrm{d}x = 2\pi \int_4^9 \left(u^{\frac{3}{2}} - 4u^{\frac{1}{2}}\right)\mathrm{d}x$

$= 2\pi \left[\frac{2}{5}u^{\frac{5}{2}} - \frac{8}{3}u^{\frac{3}{2}}\right]_4^9 = 2\pi \left[\left(\frac{486}{5} - 72\right) - \left(\frac{64}{5} - \frac{64}{3}\right)\right] = \frac{1012\pi}{15}$ units^2

8) d)

$y = \cosh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh x$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + (\sinh x)^2 = 1 + \sinh^2 x = \cosh^2 x$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{\cosh^2 x}$

$\Rightarrow x\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = x\cosh x$

$S = 2\pi \int_0^{\ln 2} x\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi \int_0^{\ln 2} x\cosh x\,\mathrm{d}x$

Use Integration by Parts:

$u = x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \cosh x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $\qquad$ $v = \sinh x$

$\therefore \int x\cosh x\,\mathrm{d}x = x\sinh x - \int \sinh x\,\mathrm{d}x$

$= x\sinh x - \cosh x + c$

So that:

$2\pi \int_0^{\ln 2} x\cosh x\,\mathrm{d}x = 2\pi[x\sinh x - \cosh x]_0^{\ln 2}$

$= 2\pi \left[\left(\frac{3}{4}\ln 2 - \frac{5}{4}\right) - (0 - 1)\right] = \frac{1}{2}\pi(3\ln 2 - 1)$ units^2

9)

$y = ax$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\therefore 1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1 + a^2$

$\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1 + a^2}$

$\Rightarrow y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = x\sqrt{1+a^2}$

$S = 2\pi\int_0^k x\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi\int_0^k x\sqrt{1+a^2}\,\mathrm{d}x$

$= 2\pi\sqrt{1+a^2}\left[\frac{1}{2}x^2\right]_0^k$

$= \pi k^2\sqrt{1+a^2}$ units^2

10)

$y = \operatorname{arcosh} x \Rightarrow \cosh y = x$

$\frac{\mathrm{d}x}{\mathrm{d}y} = \sinh y \;\; \therefore \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sinh y} = \frac{1}{\sqrt{x^2-1}}$

$\therefore 1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 1+\left(\frac{1}{\sqrt{x^2-1}}\right)^2 = 1+\frac{1}{x^2-1}$

$\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1+\frac{1}{x^2-1}} = \sqrt{\frac{x^2}{x^2-1}}$

$\Rightarrow x\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = x\sqrt{\frac{x^2}{x^2-1}} = \frac{x^2}{\sqrt{x^2-1}}$

$S = 2\pi\int_1^2 x\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = 2\pi\lim_{a\to1^+}\int_a^2 \frac{x^2}{\sqrt{x^2-1}}\mathrm{d}x$

Let $x = \cosh u$ so $\frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u\mathrm{d}$

So $x^2-1 = \cosh^2 u - 1 = \sinh^2 u$

As $a \to 1^+$, $u = 0$

when $x = 2$, $u = \operatorname{arcosh} 2$

$2\pi\lim_{a\to1^+}\int_a^2 \frac{x^2}{\sqrt{x^2-1}}\mathrm{d}x = 2\pi\lim_{a\to1^+}\int_{\operatorname{arcosh} a}^{\operatorname{arcosh} 2}\frac{\cosh^2 u}{\sqrt{\sinh^2 u}}\sinh u\,\mathrm{d}u = 2\pi\int_0^{\operatorname{arcosh} 2}\cosh^2 u\,\mathrm{d}u$

$= 2\pi\int_0^{\operatorname{arcosh} 2}\left(\frac{1}{2}+\frac{1}{2}\cosh 2u\right)\mathrm{d}u$

$= 2\pi\left[\frac{1}{2}u+\frac{1}{4}\sinh 2u\right]_0^{\operatorname{arcosh} 2} = 2\pi\left[\left(\frac{1}{2}\operatorname{arcosh} 2+\sqrt{3}\right)-(0+0)\right]$

$= 2\pi\left(\frac{1}{2}\ln\left(2+\sqrt{3}\right)+\sqrt{3}\right)$

$= \left(2\pi\sqrt{3}+\pi\ln\left(2+\sqrt{3}\right)\right)$ units^2 as req.

11) a)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 6 \Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = 36$

$\frac{\mathrm{d}y}{\mathrm{d}t} = 2 \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = 4$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 40 \Rightarrow \sqrt{40} = 2\sqrt{10}$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 2t \times 2\sqrt{10} = 4\sqrt{10}t$

$2\pi \int_0^3 y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_0^3 4\sqrt{10}t\, dt = 2\pi\left[2\sqrt{10}t^2\right]_0^3 = 36\sqrt{10}\pi$ units^2

11) b)

$\frac{dx}{dt} = 2t \Rightarrow \left(\frac{dx}{dt}\right)^2 = 4t^2$

$\frac{dy}{dt} = 4t(4t^2+1)^{-\frac{1}{2}} = \frac{4t}{\sqrt{4t^2+1}} \Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(\frac{4t}{\sqrt{4t^2+1}}\right)^2 = \frac{16t^2}{4t^2+1}$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4t^2 + \frac{16t^2}{4t^2+1} = \frac{4t^2(4t^2+1)+16t^2}{4t^2+1} = \frac{4t^2(4t^2+5)}{4t^2+1}$

$\Rightarrow \sqrt{\frac{4t^2(4t^2+5)}{4t^2+1}} = 2t\sqrt{\frac{4t^2+5}{4t^2+1}}$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{4t^2+1} \times 2t\sqrt{\frac{4t^2+5}{4t^2+1}} = 2t\sqrt{4t^2+5}$

$2\pi \int_0^1 y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_0^1 2t\sqrt{4t^2+5}\, dt = \pi \int_0^1 4t(4t^2+5)^{\frac{1}{2}}\, dt$

$= \pi\left[\frac{1}{3}(4t^2+5)^{\frac{3}{2}}\right]_0^1 = \pi\left(9 - \frac{5\sqrt{5}}{3}\right)$ units^2

12) a)

$\frac{dx}{dt} = 8t \Rightarrow \left(\frac{dx}{dt}\right)^2 = 64t^2$

$\frac{dy}{dt} = 2t \Rightarrow \left(\frac{dy}{dt}\right)^2 = 4t^2$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 64t^2 + 4t^2 = 68t^2 \Rightarrow \sqrt{68t^2} = 2\sqrt{17}t$

So $x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 4t^2 \times 2\sqrt{17}t = 8\sqrt{17}t^3$

$2\pi \int_1^2 x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_1^2 8\sqrt{17}t^3\, dt = 2\pi\left[2\sqrt{17}t^4\right]_1^2 = 2\pi\left(32\sqrt{17} - 2\sqrt{17}\right)$

$= 60\sqrt{17}\pi$ units^2

12) b)

$\frac{dx}{dt} = 2\sin t \cos t = \sin(2t) \Rightarrow \left(\frac{dx}{dt}\right)^2 = \sin^2(2t)$

$\frac{dy}{dt} = 2\cos t \sin t = \sin(2t) \Rightarrow \left(\frac{dy}{dt}\right)^2 = \sin^2(2t)$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \sin^2(2t) + \sin^2(2t) \Rightarrow \sqrt{2\sin^2(2t)} = \sqrt{2}\sin(2t)$

So $x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}=\sin^2 t\times\sqrt{2}\sin(2t)=2\sqrt{2}\sin^3 t\cos t$

$2\pi\int_0^{\frac{\pi}{2}}x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt=2\pi\int_0^{\frac{\pi}{2}}2\sqrt{2}\sin^3 t\cos t\,dt$

$=2\pi\left[\frac{\sqrt{2}}{2}\sin^4 t\right]_0^{\frac{\pi}{2}}=2\pi\left(\frac{\sqrt{2}}{2}-0\right)$

$=\sqrt{2}\pi$ units^2

13)

$\frac{dx}{d\theta}=-r\sin\theta\Rightarrow\left(\frac{dx}{d\theta}\right)^2=r^2\sin^2\theta$

$\frac{dy}{d\theta}=r\cos\theta\Rightarrow\left(\frac{dy}{d\theta}\right)^2=r^2\cos^2\theta$

$\therefore\left(\frac{dx}{d\theta}\right)^2+\left(\frac{dy}{d\theta}\right)^2=r^2\sin^2\theta+r^2\cos^2\theta=r^2(\sin^2\theta+\cos^2\theta)=r^2$

$\Rightarrow\sqrt{r^2}=r$

So $y\sqrt{\left(\frac{dx}{d\theta}\right)^2+\left(\frac{dy}{d\theta}\right)^2}=r\sin\theta\times r=r^2\sin\theta$

$2\pi\int_0^{\pi}y\sqrt{\left(\frac{dx}{d\theta}\right)^2+\left(\frac{dy}{d\theta}\right)^2}\,d\theta=2\pi\int_0^{\pi}r^2\sin\theta\,d\theta$

$=2\pi[-r^2\cos\theta]_0^{\pi}=2\pi(r^2+r^2)$

$=4\pi r^2$ units^2 as req.

14) a)

$\frac{dx}{dt}=-\tanh t\,\mathrm{sech}\,t\Rightarrow\left(\frac{dx}{dt}\right)^2=(-\tanh t\,\mathrm{sech}\,t)^2=\tanh^2 t\,\mathrm{sech}^2 t$

$\frac{dy}{dt}=\mathrm{sech}^2 t\Rightarrow\left(\frac{dy}{dt}\right)^2=(\mathrm{sech}^2 t)^2=\mathrm{sech}^4 t$

$\therefore\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2=\tanh^2 t\,\mathrm{sech}^2 t+\mathrm{sech}^4 t=\mathrm{sech}^2 t\,(\tanh^2 t+\mathrm{sech}^2 t)=\mathrm{sech}^2 t$

14) b)

$\sqrt{\mathrm{sech}^2 t}=\mathrm{sech}\,t$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}=\tan t\times\mathrm{sech}\,t=\tanh t\,\mathrm{sech}\,t$

$2\pi\int_0^{\ln 5}y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt=2\pi\int_0^{\ln 5}\tanh t\,\mathrm{sech}\,t\,dt$

$=2\pi[-\mathrm{sech}\,t]_0^{\ln 5}=2\pi\left(-\frac{5}{13}--1\right)$

$=\frac{16\pi}{13}$ units^2

15) a)

$\frac{dx}{dt} = 6(1+t)^{\frac{1}{2}} \Rightarrow \left(\frac{dx}{dt}\right)^2 = \left(6(1+t)^{\frac{1}{2}}\right)^2 = 36(1+t)$

$\frac{dy}{dt} = -6(1-t)^{\frac{1}{2}} \Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(-6(1-t)^{\frac{1}{2}}\right)^2 = 36(1-t)$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 36(1+t) + 36(1-t) = 72$

$\Rightarrow \sqrt{72} = 6\sqrt{2}$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 4(1-t)^{\frac{3}{2}} \times 6\sqrt{2} = 24\sqrt{2}(1-t)^{\frac{3}{2}}$

$2\pi \int_0^{0.8} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_0^{0.8} \left(24\sqrt{2}(1-t)^{\frac{3}{2}}\right) dt$

$= 2\pi \left[-\frac{48}{5}\sqrt{2}(1-t)^{\frac{5}{2}}\right]_0^{0.8}$

$= 2\pi \left(-\frac{48}{5}\sqrt{2}(0.2)^{\frac{5}{2}} - -\frac{48}{5}\sqrt{2}\right)$

$= 83.8$ cm^2 (3 SF)

15) b)

$15 \times 83.77740 \ldots \times 0.01 = £25$ (nearest £1)

16)

$\frac{dx}{dt} = -3k \sin t \cos^2 t \Rightarrow \left(\frac{dx}{dt}\right)^2 = (-3k \sin t \cos^2 t)^2 = 9k^2 \sin^2 t \cos^4 t$

$\frac{dy}{dt} = 3k \cos t \sin^2 t \Rightarrow \left(\frac{dy}{dt}\right)^2 = (3k \cos t \sin^2 t)^2 = 9k^2 \cos^2 t \sin^4 t$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9k^2 \sin^2 t \cos^4 t + 9k^2 \cos^2 t \sin^4 t$

$= 9k^2 \sin^2 t \cos^2 t (\cos^2 t + \sin^2 t) = 9k^2 \sin^2 t \cos^2 t$

$\Rightarrow \sqrt{9k^2 \sin^2 t \cos^2 t} = 3k \sin t \cos t$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = k \sin^3 t \times 3k \sin t \cos t = 3k^2 \sin^4 t \cos t$

$2\pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (3k^2 \sin^4 t \cos t)\, dt$

$= 2\pi k^2 \left[\frac{3}{5}\sin^5 t\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 2\pi k^2 \left(\frac{3}{5} - \frac{3\sqrt{2}}{40}\right) = \frac{3}{20}\pi k^2 (8 - \sqrt{2})$ units^2 as req.

17)

$\frac{dx}{dt} = 2 \Rightarrow \left(\frac{dx}{dt}\right)^2 = 4$

$\frac{dy}{dt} = t - \frac{1}{t} \Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(t - \frac{1}{t}\right)^2 = t^2 - 2 + \frac{1}{t^2}$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4 + t^2 - 2 + \frac{1}{t^2} = t^2 + 2 + \frac{1}{t^2} = \left(t + \frac{1}{t}\right)^2$

$\Rightarrow \sqrt{\left(t + \frac{1}{t}\right)^2} = t + \frac{1}{t}$

So $x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 2t \times \left(t + \frac{1}{t}\right) = 2t^2 + 2$

$2\pi \int_1^2 x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_1^2 (2t^2 + 2)\, dt$

$= 2\pi \left[\frac{2}{3}t^3 + 2t\right]_1^2 = 2\pi \left(\left(\frac{16}{3} + 4\right) - \left(\frac{2}{3} + 2\right)\right)$

$= \frac{40}{3}\pi$ units^2

18) a)

Use Integration by Parts:

$u = \ln x \qquad \frac{dv}{dx} = \frac{1}{x}$

$\frac{du}{dx} = \frac{1}{x} \qquad v = \ln x$

$\therefore \int \frac{\ln x}{x} dx = \ln^2 x - \int \frac{\ln x}{x} dx$

So $2\int \frac{\ln x}{x} dx = \ln^2 x$

$\therefore \int \frac{\ln x}{x} dx = \frac{1}{2}\ln^2 x + c$ as req.

18) b)

$\frac{dx}{dt} = 2 \Rightarrow \left(\frac{dx}{dt}\right)^2 = 4$

$\frac{dy}{dt} = t - \frac{1}{t} \Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(t - \frac{1}{t}\right)^2 = t^2 - 2 + \frac{1}{t^2}$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4 + t^2 - 2 + \frac{1}{t^2} = t^2 + 2 + \frac{1}{t^2} = \left(t + \frac{1}{t}\right)^2$

$\Rightarrow \sqrt{\left(t + \frac{1}{t}\right)^2} = t + \frac{1}{t}$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \left(\frac{1}{2}t^2 - \ln t\right) \times \left(t + \frac{1}{t}\right) = \frac{1}{2}t^3 - t\ln t + \frac{1}{2}t - \frac{\ln t}{t}$

$2\pi \int_1^3 y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_1^3 \left(\frac{1}{2}t^3 - t\ln t + \frac{1}{2}t - \frac{\ln t}{t}\right) dt$

Consider $\int t \ln t \, \mathrm{d}t$:

Use Integration by Parts:

$$u = \ln t \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = t$$
$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{1}{t} \qquad v = \frac{1}{2}t^2$$

$\therefore \int t \ln t \, \mathrm{d}t = \frac{1}{2}t^2 \ln t - \frac{1}{2}\int t \, \mathrm{d}x = \frac{1}{2}t^2 \ln t - \frac{1}{4}t^2 + c$

So that

$2\pi \int_1^3 \left(\frac{1}{2}t^3 - t \ln t + \frac{1}{2}t - \frac{\ln t}{t}\right) \mathrm{d}t$

$= 2\pi \left[\frac{1}{8}t^4 - \frac{1}{2}t^2 \ln t + \frac{1}{4}t^2 + \frac{1}{4}t^2 - \frac{1}{2}\ln^2 t\right]_1^3$

$= 2\pi \left(\left(\frac{81}{8} - \frac{9}{2}\ln 3 + \frac{9}{2} - \frac{1}{2}\ln^2 3\right) - \left(\frac{1}{8} - 0 + \frac{1}{2} - 0\right)\right)$

$= 53.1 \text{ units}^2 \text{ (3 SF)}$

19)

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \mathrm{e}^\theta \sin\theta + \mathrm{e}^\theta \cos\theta = \mathrm{e}^\theta(\sin\theta + \cos\theta) \Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 = \mathrm{e}^{2\theta}(\sin\theta + \cos\theta)^2$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \mathrm{e}^\theta \cos\theta - \mathrm{e}^\theta \sin\theta = \mathrm{e}^\theta(\cos\theta - \sin\theta) \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2 = \mathrm{e}^{2\theta}(\cos\theta - \sin\theta)^2$

$\therefore \left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2 = \mathrm{e}^{2\theta}(\sin\theta + \cos\theta)^2 + \mathrm{e}^{2\theta}(\cos\theta - \sin\theta)^2$

$= \mathrm{e}^{2\theta}((\sin\theta + \cos\theta)^2 + (\cos\theta - \sin\theta)^2)$

$= \mathrm{e}^{2\theta}(\sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta + \cos^2\theta - 2\sin\theta\cos\theta + \sin^2\theta)$

$= \mathrm{e}^{2\theta}(2\sin^2\theta + 2\cos^2\theta)$

$= 2\mathrm{e}^{2\theta}$

$\Rightarrow \sqrt{2\mathrm{e}^{2\theta}} = \sqrt{2}\mathrm{e}^\theta$

So $y\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2} = \mathrm{e}^\theta \cos\theta \times \sqrt{2}\mathrm{e}^\theta = \sqrt{2}\mathrm{e}^{2\theta}\cos\theta$

$2\pi \int_0^{\frac{\pi}{2}} y\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2} \, \mathrm{d}\theta = 2\pi \int_0^{\frac{\pi}{2}} \left(\sqrt{2}\mathrm{e}^{2\theta}\cos\theta\right) \mathrm{d}\theta$

Consider $\int \mathrm{e}^{2\theta}\cos\theta \, \mathrm{d}\theta$:

Use Integration by Parts:

$$u = \mathrm{e}^{2\theta} \qquad \frac{\mathrm{d}v}{\mathrm{d}\theta} = \cos\theta$$
$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = 2\mathrm{e}^{2\theta} \qquad v = \sin\theta$$

$\therefore \int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta - 2\int e^{2\theta}\sin\theta\,d\theta$

Use Integration by Parts:

$u = e^{2\theta}$ $\quad\quad$ $\frac{dv}{d\theta} = \sin\theta$

$\frac{du}{d\theta} = 2e^{2\theta}$ $\quad\quad$ $v = -\cos\theta$

$\therefore \int e^{2\theta}\sin\theta\,d\theta = -e^{2\theta}\cos\theta + 2\int e^{2\theta}\cos\theta\,d\theta$

So that

$\int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta - 2\int e^{2\theta}\sin\theta\,d\theta = e^{2\theta}\sin\theta - 2\left(-e^{2\theta}\cos\theta + 2\int e^{2\theta}\cos\theta\,d\theta\right)$

$\Rightarrow \int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta - 4\int e^{2\theta}\cos\theta\,d\theta$

$\Rightarrow 5\int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta$

$\Rightarrow \int e^{2\theta}\cos\theta\,d\theta = \frac{1}{5}e^{2\theta}\sin\theta + \frac{2}{5}e^{2\theta}\cos\theta + c$

Use this result:

$2\pi\int_0^{\frac{\pi}{2}}\left(\sqrt{2}e^{2\theta}\cos\theta\right)d\theta = 2\sqrt{2}\pi\left[\frac{1}{5}e^{2\theta}\sin\theta + \frac{2}{5}e^{2\theta}\cos\theta\right]_0^{\frac{\pi}{2}}$

$= 2\sqrt{2}\pi\left(\left(\frac{1}{5}e^{\pi} + 0\right) - \left(0 + \frac{2}{5}\right)\right)$

$= \frac{2\sqrt{2}\pi}{5}(e^{\pi} - 2)$ units^2 as req.

20) a)

Trial solution:

$\int e^{-2x}(\cos x - \sin x)dx = e^{-2x}(A\cos x + B\sin x)$

$\frac{d}{dx}[e^{-2x}(A\cos x + B\sin x)] = -2e^{-2x}(A\cos x + B\sin x) + e^{-2x}(-A\sin x + B\cos x)$

$= e^{-2x}\left((-2A + B)\cos x + (-A - 2B)\sin x\right)$

So $-2A + B = 1$ and $-A - 2B = -1$

Solving simultaneously gives $A = -\frac{1}{5}$ and $B = \frac{3}{5}$

$\therefore \int e^{-2x}(\cos x - \sin x)dx = e^{-2x}\left(-\frac{1}{5}\cos x + \frac{3}{5}\sin x\right) = -\frac{1}{5}e^{-2x}(\cos x - 3\sin x)$ as req.

20) b)

$\frac{dx}{dt} = e^{-t}(\cos t - \sin t) - e^{-t}(\cos t + \sin t) = -2e^{-t}\sin t$

$\Rightarrow \left(\frac{dx}{dt}\right)^2 = (-2e^{-t}\sin t)^2 = 4e^{-2t}\sin^2 t$

$\frac{dy}{dt} = -e^{-t}(\cos t - \sin t) + e^{-t}(-\sin t - \cos t) = -2e^{-t}\cos t$

$\Rightarrow \left(\frac{dy}{dt}\right)^2 = (-2e^{-t}\cos t)^2 = 4e^{-2t}\cos^2 t$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4e^{-2t}\sin^2 t + 4e^{-2t}\cos^2 t = 4e^{-2t}(\sin^2 t + \cos^2 t) = 4e^{-2t}$

$\Rightarrow \sqrt{4e^{-2t}} = 2e^{-t}$

So $y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \left(e^{-t}(\cos t - \sin t)\right) \times 2e^{-t} = 2e^{-2t}(\cos t - \sin t)$

$2\pi \int_0^{\frac{\pi}{2}} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_0^{\frac{\pi}{2}} (2e^{-2t}\cos t - 2e^{-2t}\sin t)\, dt$

Since $\int e^{-2x}(\cos x - \sin x)dx = -\frac{1}{5}e^{-2x}(\cos x - 3\sin x)$ then

$2\pi \int_0^{\frac{\pi}{2}} (2e^{-2t}\cos t - 2e^{-2t}\sin t)\, dt = 4\pi \left[-\frac{1}{5}e^{-2t}(\cos t - 3\sin t)\right]_0^{\frac{\pi}{2}}$

$= 4\pi\left(\left(\frac{3}{5}e^{-\pi}\right) - \left(-\frac{1}{5}\right)\right)$

$= \frac{4\pi}{5}\left(1 + \frac{3}{e^{\pi}}\right)$ units^2